Bioacoustic Drought Detection: Deliverable 4 Update

To the group, from Thabiso Masenya · 17 October 2026 · A full explanation document (extended *Deliverable 4 Explained*) will follow.

Where we stand. Deliverable 4 (Implementation) is built, tested and written up. The full pipeline runs end to end with one command. All experiments are finished, and the report covers all 8 rubric sections. Our model identifies drought stress from a single 2 ms plant click about **7 times in 10** on plants it has never heard, which matches the original scientists' result.

What was built

- **A complete, automated pipeline** (Python, TensorFlow): it checks and fingerprints all 5,477 sound files, filters them to 20–100 kHz, splits the data *by plant*, mixes real greenhouse noise into training clips, trains, tests and produces charts and HTML reports.
- **Four models compared fairly** on identical data: our 1D CNN (reads the raw sound wave), the 2D Mel-CNN and LSTM from Deliverable 1, and the 1D CNN without noise training.
- **Professional practice:** one settings file, 34 automatic tests, git version history, and results that can be reproduced exactly from saved settings.

Key results (5-fold cross-validation, every plant tested once)

Model	Balanced accuracy	ROC-AUC	What it means
1D CNN + noise training (ours)	**70.1%**	**0.781**	Best on every measure
2D Mel-CNN (Deliverable 1 plan)	68.4%	0.757	Close second; the gap is **not** statistically proven
Mel-LSTM (Deliverable 1 plan)	64.2%	0.705	Clearly worse (proven)
Always guess "dry"	50.0%	0.500	The floor any model must beat

Noise training makes no difference on clean sound, but helps when it gets noisy: the lead grows from +0.030 to +0.044 AUC when greenhouse noise is as loud as the click. The model works well on **tomato** (AUC 0.81) but poorly on **tobacco** (0.59).

What changed from Deliverable 1, and why

- **Mel-spectrogram to raw-wave 1D CNN:** a 2 ms click is too short for a spectrogram (it gets blurred), and the Mel scale is designed for human hearing. We kept the old designs to test the change.
- **Random split to split by plant:** one plant makes up to 148 similar clicks, so a random split lets the model "cheat" by recognising plants it has already heard.
- **Tomato only to tomato + tobacco:** 34% more data, with results reported per species.
- **Dropped:** the three severity levels and the early-warning time (Δt). The dataset only says cut or dry, and has no record of when plants wilted.

Bottlenecks we hit, and how they were resolved

Bottleneck	Resolution
The starting code was only the model, and it came from the original authors; 35 of the 50 marks depended on the missing parts	Built the rest (about 2,000 lines, 13 modules); credited their code openly
The first test split held only 2 cut-tomato plants, so the score was unreliable (range 0.62–0.89)	The splitter now picks the most balanced of 40 splits; tuning was re-run to avoid leakage
The test set preferred the *old* 2D design (0.729 vs 0.692)	No tweaking on the test set; a fair 5-fold comparison showed ours ahead but not significantly, so we changed our claim
Slow compute (CPU only; the full run takes 3–4 h), and a cloud restart lost a job	Long stages now save progress and resume; a re-run reproduced the identical score
Data limits: tobacco near chance; no Δt or severity labels	Reported honestly as limitations, with next steps

Three messages we must all get right when presenting

- Say **"at least as good as the spectrogram model"**, never "superior" or "significantly better". The gap over the 2D CNN is not statistically significant ($p = 0.27$).
- Always say **"on plants it has never heard"**: that is what makes our 70% a fair score.
- **Ready for further testing and a supervised tomato trial, not for automatic irrigation.**

Next steps for the group

- Read the full explanation document when I send it. It defines every term and has a slide plan and 18 likely lecturer questions with answers.
- Each person picks a presentation part: introduction and data, preprocessing, model and pipeline, or results and conclusion.
- Send me any question you can't answer yet, so I can cover it in the extended document.